Perspective-Conditioned Geopolitical Event Prediction

Complete Architecture Specification

Abstract. This document gives the complete, confirmed specification of the Lancer architecture. The model runs 100 independent *mini models*, each of which takes four natural-language strings, embeds them through MiniLM-v2 and custom 64-d converters, and produces a binary probability vector $[\hat{p}^+, \hat{p}^-]$. An aggregation head then pools all 100 probability votes and 100 personality vectors into a single final prediction $[\hat{y}^+, \hat{y}^-]$.

Part 1 — The Mini Model (runs 100 times independently)

Each of the 100 mini models receives four raw strings and produces one two-number probability vector. Every mini model has its own independent weights. No parameters are shared between mini models.

Step 1 — String inputs and embedding

The four inputs are plain English strings:

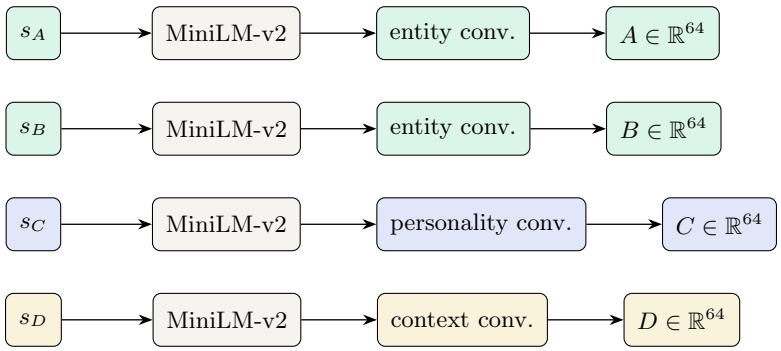
Input	Example	Space
s_A	*"United States"*	entity (shared with B)
s_B	*"Iran"*	entity (shared with A)
s_C	*"realist military analyst"*	personality
s_D	*"Gulf War, 1990, Iraq invades Kuwait"*	context

Each string passes through the two-step embedding pipeline:

$$s \xrightarrow{\text{MiniLM-v2}} \mathbf{z} \in \mathbb{R}^{384} \xrightarrow{\text{converter}_k} \mathbf{e} \in \mathbb{R}^{64}$$

MiniLM-v2 is a frozen pretrained sentence transformer. Each converter_k is a learned two-layer

MLP trained with triplet loss to map the 384-d MiniLM output into a custom 64-d space:



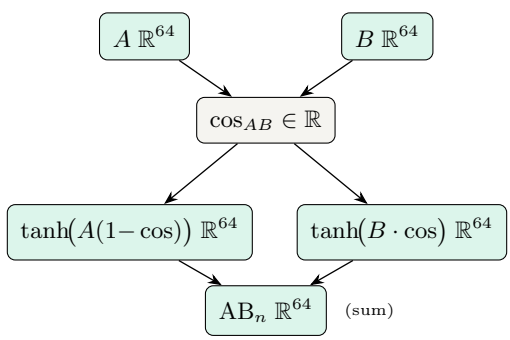
Entity converter is shared. s_A and s_B both pass through the same entity converter, which is why cosine similarity between A and B is geometrically meaningful.

Step 2 — Actor–receiver blend (AB_n)

We measure how similar A and B are in the shared embedding space, then blend them:

$$\cos_{AB} = \frac{A \cdot B}{\|A\| \, \|B\|} \in [-1, 1]$$

$$\mathrm{AB}_n = \underbrace{\tanh\big(A \cdot (1 - \cos_{AB})\big)}_{\text{actor dominates when } A,B \text{ differ}} + \underbrace{\tanh\big(B \cdot \cos_{AB}\big)}_{\text{receiver dominates when } A,B \text{ agree}} \in \mathbb{R}^{64}$$

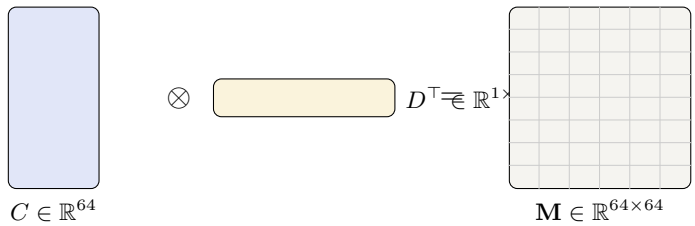


Step 3 — Personality–context outer product matrix

The personality vector C and context vector D are combined via outer product to produce a 64×64 matrix:

$$\mathbf{M} = C \otimes D \in \mathbb{R}^{64 \times 64}$$

Flattened to a vector of 4096 values, the full matrix is:



Step 4 — Residual refinement ($\times 3$)

Three passes refine $\mathbf{M}$. Each pass applies a shared scalar α and a per-cell bias $\mathbf{B}$, then adds the previous state back:

$$\mathbf{M}^{(t)} \;=\; (1+\alpha)\,\mathbf{M}^{(t-1)} \;+\; \mathbf{B} \qquad t = 1, 2, 3$$

64×64; [mat,right=of m0] (m1) $\mathbf{M}^{(1)}$
64×64; [mat,right=of m1] (m2) $\mathbf{M}^{(2)}$
64×64; [mat,right=of m2,fill=intuitbg,draw=intuitborder] (m3) $\mathbf{M}'$

64×64; [arr] (m0)–(m1) node[midway,above,font=](1+α) $\cdot$ +$\mathbf{B}$; [arr] (m1)–(m2) node[midway,above,font=](1+α) $\cdot$ +$\mathbf{B}$; [arr] (m2)–(m3)

node[midway,above,font=](1+α) $\cdot$ +$\mathbf{B}$;

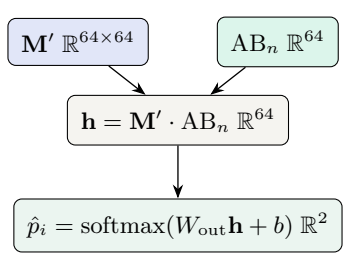
Step 5 — Collapse to probability via ABn

The refined matrix $\mathbf{M}' \in \mathbb{R}^{64\times 64}$ is multiplied by $\mathrm{AB}_n \in \mathbb{R}^{64}$ to yield a 64-dimensional vector, which is then projected and softmax'd to give the mini model's binary probability output:

$$\mathbf{h} \;=\; \mathbf{M}' \cdot \mathrm{AB}_n \;\in\; \mathbb{R}^{64}$$

$$\hat{p}_i \;=\; \mathrm{softmax}\big(W_{\text{out}}^{(i)}\,\mathbf{h} + b_{\text{out}}^{(i)}\big) \;\in\; \mathbb{R}^2$$

where $W_{\text{out}}^{(i)} \in \mathbb{R}^{2\times 64}$ and $b_{\text{out}}^{(i)} \in \mathbb{R}^2$ are unique to mini-model i.



Mini model output: $\hat{p}_i = [\hat{p}_i^+,\ \hat{p}_i^-] \in \mathbb{R}^2, \quad \hat{p}_i^+ + \hat{p}_i^- = 1$

This runs 100 times. Each mini model i sees its own $(s_A,\ s_B,\ s_{C_i},\ s_{D_i})$ and produces its own independent $\hat{p}_i$. No weights shared between mini models.

Part 2 — Aggregation Head

After all 100 mini models have run, the aggregation head combines their probability votes with their personality vectors to produce the final prediction.

Step 6 — Personality sharpening and projection (C'_n)

Each of the 100 personality vectors $C_i \in \mathbb{R}^{64}$ is processed independently:

(a) Scale and softmax:

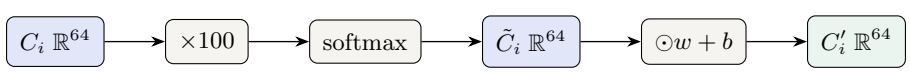
$$\tilde{C}_i \;=\; \text{softmax}(100 \cdot C_i) \;\in\; \mathbb{R}^{64}$$

Multiplying by 100 sharpens the distribution before normalising, making dominant personality dimensions stand out clearly.

(b) Pairwise learned weights and bias:

$$C'_i \;=\; w \;\odot\; \tilde{C}_i \;+\; b \;\in\; \mathbb{R}^{64}$$

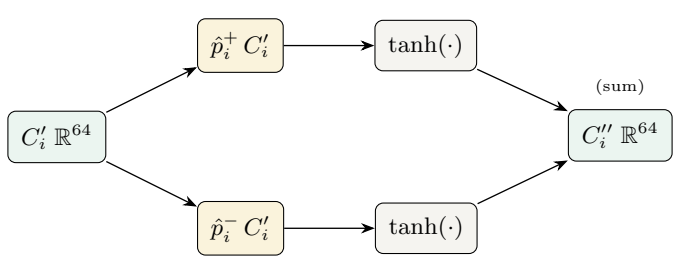
where $w, b \in \mathbb{R}^{64}$ are learned parameters shared across all 100 personalities ($\odot$ is element-wise multiplication).



Step 7 — Confidence gating (C''_i)

Two identical copies of C'_i are made. One is scaled by $\hat{p}_i^+$ (the mini model's positive probability), the other by $\hat{p}_i^-$ (the negative probability). Tanh is applied to each and the results are summed:

$$C''_i \;=\; \tanh\!\big(\hat{p}_i^+\, C'_i\big) \;+\; \tanh\!\big(\hat{p}_i^-\, C'_i\big) \;\in\; \mathbb{R}^{64}$$

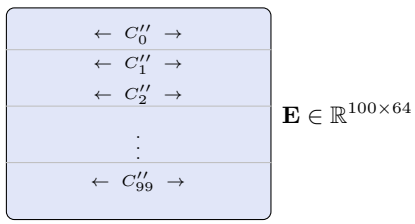


Intuition. A mini model that is very confident (one of $\hat{p}^{\pm}$ near 1) pushes C_i'' hard toward the tanh boundary — a loud, decisive vote. A mini model that is uncertain (both near 0.5) produces a soft, muted C_i'' — a quiet, hedged vote. No explicit attention mechanism is needed; confidence automatically modulates volume.

Step 8 — Build the ensemble matrix E

All 100 confidence-gated vectors are stacked row-wise:

$$\mathbf{E} \;=\; \begin{pmatrix} \longleftarrow C_0'' \longrightarrow \\ \longleftarrow C_1'' \longrightarrow \\ \vdots \\ \longleftarrow C_{99}'' \longrightarrow \end{pmatrix} \;\in\; \mathbb{R}^{100\times 64}$$

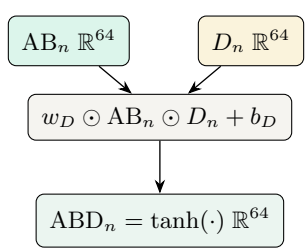


Step 9 — Build the context-fused situation vector (ABD_n)

The actor-receiver blend AB_n is fused with the context vector D_n through a learned element-wise transform:

$$\mathrm{ABD}_n \;=\; \tanh\Big(w_D \;\odot\; \mathrm{AB}_n \;\odot\; D_n \;+\; b_D\Big) \;\in\; \mathbb{R}^{64}$$

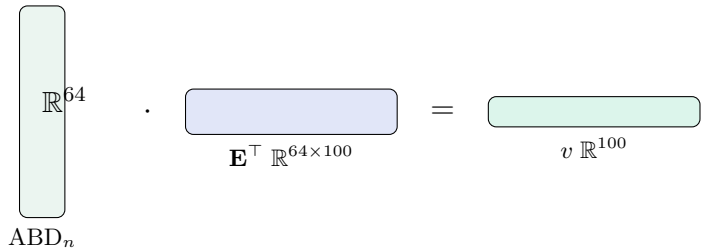
where w_D, $b_D \in \mathbb{R}^{64}$ are learned parameters. Then a weighted sum collapses everything to a single vector.



Step 10 — Poll the ensemble

$\mathrm{ABD}_n \in \mathbb{R}^{64}$ is dotted against $\mathbf{E}$ to produce a score for each of the 100 personality perspectives:

$$v \;=\; \mathrm{ABD}_n \;\cdot\; \mathbf{E}^{\top} \;\in\; \mathbb{R}^{100}$$



Element v_i measures how well the current situation aligns with personality i's confident view of the world.

Step 11 — Bottleneck refinement ($\times 3$)

Three expand–contract passes force the model to find the most information-dense summary of the 100 votes:

$$v^{(t)} \;=\; W_{\downarrow}^{(t)}\,\mathrm{ReLU}\big(W_{\uparrow}^{(t)}\,v^{(t-1)}\big)$$

$$v^{(t-1)}\ \mathbb{R}^{100} \xrightarrow{W_{\uparrow}\ \mathbb{R}^{128\times 100}} \mathbb{R}^{128} \xrightarrow{\mathrm{ReLU}} \mathbb{R}^{128} \xrightarrow{W_{\downarrow}\ \mathbb{R}^{100\times 128}} v^{(t)}\ \mathbb{R}^{100}$$

Applied three times from $v^{(0)} = v$, yielding $v^{(3)}$. Each pass has its own independent $W_{\uparrow}^{(t)}$ and $W_{\downarrow}^{(t)}$.

Step 12 — Final output

$$\hat{y} \;=\; \mathrm{softmax}\big(W_{\mathrm{out}}\,v^{(3)} + b_{\mathrm{out}}\big) \;\in\; \Delta^1$$
$$\hat{y} \;=\; [\,\hat{y}^+,\ \hat{y}^-\,], \qquad \hat{y}^+ + \hat{y}^- = 1$$

$W_{\mathrm{out}} \in \mathbb{R}^{2\times 100}$. $\hat{y}^+$ is the probability entity A takes the action toward B. $\hat{y}^-$ is the probability it does not.

End-to-End Data Flow

Inside each mini model i (runs 100 times):

$$(s_A, s_B, s_{C_i}, s_{D_i}) \xrightarrow{\text{MiniLM+conv}} A, B, C_i, D_i \in \mathbb{R}^{64}$$

$$\xrightarrow{\cos_{AB}\ \text{blend}} \mathrm{AB}_n \in \mathbb{R}^{64} \quad \xrightarrow{C_i \otimes D_i} \mathbf{M} \in \mathbb{R}^{64\times 64}$$

$$\xrightarrow{\times 3 \text{ residual}} \mathbf{M}' \in \mathbb{R}^{64\times 64} \xrightarrow{\mathbf{M}'\cdot \mathrm{AB}_n} \mathbf{h} \in \mathbb{R}^{64} \xrightarrow{W_{\text{out}},\text{softmax}} \hat{p}_i \in \mathbb{R}^2$$

Aggregation head (runs once):

$$\{C_i\}_{i=0}^{99},\ \{\hat{p}_i\}_{i=0}^{99} \xrightarrow{\times 100,\,\text{softmax},\,\odot w+b} \{C_i'\} \xrightarrow{\text{confidence gate}} \{C_i''\} \xrightarrow{\text{stack}} \mathbf{E} \in \mathbb{R}^{100\times 64}$$

$$\mathrm{AB}_n,\ D_n \xrightarrow{\odot w_D+b_D,\,\tanh} \mathrm{ABD}_n \in \mathbb{R}^{64} \xrightarrow{\cdot\,\mathbf{E}^\top} v \in \mathbb{R}^{100}$$

$$\xrightarrow{\times 3 \text{ bottleneck}} v^{(3)} \in \mathbb{R}^{100} \xrightarrow{W_{\text{out}},\,\text{softmax}} \hat{y} \in \Delta^1$$

Notation Reference

Symbol	Meaning
s_A, s_B, s_C, s_D	Raw input strings
$A, B \in \mathbb{R}^{64}$	Actor and receiver embeddings, shared space
$C_i \in \mathbb{R}^{64}$	Personality embedding for mini-model i
$D_i \in \mathbb{R}^{64}$	Context embedding for mini-model i
$\cos_{AB}$	Cosine similarity of A and B
$\mathrm{AB}_n \in \mathbb{R}^{64}$	Blended actor–receiver vector
$\mathbf{M} \in \mathbb{R}^{64\times 64}$	Outer product matrix $C_i \otimes D_i$
$\mathbf{M}'$	$\mathbf{M}$ after 3 residual passes
α	Shared scalar residual weight
$\mathbf{B} \in \mathbb{R}^{64\times 64}$	Per-cell learned bias matrix
$\mathbf{h} \in \mathbb{R}^{64}$	$\mathbf{M}' \cdot \mathrm{AB}_n$, mini-model hidden state
$\hat{p}_i \in \mathbb{R}^2$	Mini-model i probability output
$\tilde{C}_i \in \mathbb{R}^{64}$	Sharpened personality (after $\times 100$, softmax)
$C'_i \in \mathbb{R}^{64}$	After pairwise weights and bias
$C''_i \in \mathbb{R}^{64}$	Confidence-gated personality
$\mathbf{E} \in \mathbb{R}^{100\times 64}$	Stacked ensemble matrix
$\mathrm{ABD}_n \in \mathbb{R}^{64}$	Context-fused situation vector
$v \in \mathbb{R}^{100}$	Poll vector (dot of ABD_n and $\mathbf{E}^\top$)
$v^{(3)} \in \mathbb{R}^{100}$	Refined poll after 3 bottleneck passes
$\hat{y} \in \Delta^1$	Final binary probability output
$\mathrm{converter}_k$	MiniLM $\rightarrow$ 64-d learned MLP, space k
$\odot$	Element-wise (Hadamard) product
$\otimes$	Outer product